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ARTICLE
Animal Ecology

An inverse latitudinal gradient in intraspecific genetic diversity of a dominant ant in the neotropical savanna

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Abstract

Although latitudinal gradient of diversity (LGD) is a biogeographic pattern widely shown for distinct animal groups, some display the opposite pattern, with biodiversity peaking away from the equator. These inverse gradients may result from drivers that also overlap with extrinsic factors shaping genetic diversity across taxa, a level of biodiversity less investigated under the LGD lens. Recent studies have reported an inverse trend of LGD for ant species and food web diversity in the Neotropical Cerrado savanna. However, it is unclear whether latitudinal gradients in diversity translate across biological scales to the most fundamental level of biodiversity: intraspecific genetic diversity (iGD). Here, we investigated the latitudinal distribution of iGD in one of the most abundant ants in Cerrado: *Camponotus crassus* (Formicidae: Formicinae). We sampled workers from distinct Cerrado localities distributed across a latitudinal gradient and asked whether latitude, or any relevant abiotic and/or biotic factor, explained iGD. Workers were genotyped for single nucleotide polymorphisms, and iGD was evaluated against data for climate, ant community, availability of sugar-rich resources, local abundance, and geographic distribution of *C. crassus*. Consistent with diversity across other scales, all iGD estimates increased significantly with latitude. iGD was negatively correlated with annual precipitation, potentially due to its role in regulating

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nuptial flights in ants. iGD also showed a positive correlation with sugar-rich food availability, suggesting that access to key resources may influence demographic and evolutionary processes in ant populations. Our large-scale work fills a knowledge gap of the patterns and drivers of ant diversity across dry and warm ecosystems in a dominant ant species. It suggests that an inverse LGD for ants is not restricted to species and ecological interaction levels, but also translates to the intraspecific level, adding a novel perspective to the current knowledge of ant biodiversity distribution in the Neotropics.

KEYWORDS
ant–plant interactions, *Camponotus*, Cerrado, Formicidae, genetic diversity, Hymenoptera, latitudinal gradient of biodiversity, Neotropics, single nucleotide polymorphism

INTRODUCTION

One of the most consistent patterns of life distribution on Earth is the latitudinal gradient of diversity (LGD), according to which biodiversity increases toward the tropics under lower latitudes (Gaston, 2000; Pianka, 1966; Rosenzweig, 1995; Willig et al., 2003). Although biodiversity can be defined as the variety of life at all biological scales, from genes to ecosystems (Colwell, 2009), most studies on LGD focus on species richness (Gaston, 2000; Hillebrand, 2004; Willig et al., 2003). Thus, evidence on global intraspecific genetic diversity (iGD) distribution patterns is still lacking, especially in warm regions (see De Kort et al., 2021; Pereira, 2016). Although distinct animal lineages have a higher diversity at lower latitudes, some groups may display an inverse pattern (Cerezer et al., 2022). Despite highly taxon-specific and no universal explanation (Cerezer et al., 2022), inverse LGDs (i.e., accumulation of more biodiversity away from equator) can result from the very same drivers of traditional LGD: longer time of colonization, intrinsic properties (such as temperature, topography, seasonality) or higher availability of energy, niches and resources at high latitude regions (Mittelbach et al., 2007). Interestingly, these mechanisms directly overlap with extrinsic factors known to influence genetic diversity across taxa (e.g., Pamilo, 1988; Vellend & Geber, 2005).

Genetic variation is associated with environmental descriptors both at local and global scales (Hedrick, 1986; Pamilo, 1988; Stein et al., 2014; Stein & Kreft, 2015; Vellend & Geber, 2005). Habitat complexity increases the chances that species can exploit resources in variable modes (Tews et al., 2004), representing a potential driver for differential selection (Vellend & Geber, 2005). Thus, environmental gradients may influence genetic variation directly through natural selection (the fitness of distinct genotypes may potentially vary under specific microhabitat conditions) or indirectly, with environmental variation affecting population demographic processes that shape genetic diversity (Pamilo, 1988). In most terrestrial habitats, physical environmental structure is mainly promoted by vegetation, which affects animal diversity at distinct biological scales (Tews et al., 2004). Vegetation structure includes characteristics such as plant density and diversity (Stein & Kreft, 2015), which are often correlated (Qian & Kissling, 2010). In addition to vegetation, terrestrial environments can also be distinguished by macro and microclimate (Hillebrand, 2004; Stein & Kreft, 2015), both of which may play a role in intraspecific genetic variation.

iGD is also commonly correlated with species community diversity for different reasons (Martin & McKay, 2004; Pereira, 2016; Vellend & Geber, 2005). First, both these levels of diversity are subject to similar evolutionary forces: (1) drift, which can lead to random variations in abundance of alleles or species, (2) migration, which can introduce new alleles to populations or new species to communities, and (3) selection, which may favor individuals from distinct species or genotypes from the same species (see Vellend & Geber, 2005). Additionally, genetic diversity may impact species diversity if intraspecific genotypic variation is translated into phenotypic variation, with ecological effects on other species (Hughes et al., 2008). Moreover, increasing genotype variation can enable species to respond differentially to selective pressure from competition, promoting species coexistence (Vellend, 2006) and the emergence of new phenotypes that can efficiently use unexplored resources (Jousset et al., 2016; Vellend & Geber, 2005).

Finally, genetic variation can be directly associated with resource availability in the environment, given the potential role of food resources in mediating demography and life history features such as survivorship and reproduction (e.g., Tavecchia et al., 2007).

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Ants in the Cerrado savanna and inverse LGD across multiple biodiversity levels

Cerrado covers more than two million km^2, nearly 26% of Brazilian territory (Figure 1A), comprising a mosaic of vegetation types (Oliveira-Filho & Ratter, 2002; Vieira et al., 2022) and is considered one of the world's biodiversity hotspots (Myers et al., 2000). Ants are ubiquitous in this biome, both on ground and foliage (Brandão et al., 2011; Schoereder et al., 2010; Vasconcelos et al., 2018, 2023). Among other tropical savannas, Cerrado has the highest arboreal ant species diversity (Campos et al., 2011). The dominance of ants in Cerrado vegetation is promoted by the presence of nesting sites on plants (e.g., hollowed stems) and a high incidence of liquid food sources on leaves from extrafloral nectaries (EFNs) and trophobiont insects, such as honeydew-producing hemipterans and some lepidopteran larvae (Kaminski et al., 2010; Oliveira & Freitas, 2004; Oliveira & Leitao-Filho, 1987). Moreover, the spatial heterogeneity promoted by the vegetation mosaic and its vertical stratification contribute to the high number of ant species in the Cerrado (Ribas et al., 2003). Based on extensive data collection across the Cerrado domain, researchers have found a positive association between latitude and: (1) ant species richness (Vasconcelos et al., 2018), (2) ecological interaction networks, (3) interaction diversity/dissimilarity (Dáttilo & Vasconcelos, 2019), (4) vertical stratification of ant assemblages (Vasconcelos et al., 2023), and (5) functional and phylogenetic ant richness (Neves et al., 2024). In other words, all these levels of ant biodiversity increase further away from the equator. However, it is unclear whether such inverse latitudinal gradients in ant diversity translate across biological scales to the most fundamental level of biodiversity: the iGD. Given the great geographic variation in environmental descriptors, species communities, and food resources availability (Figure 1B), the Cerrado biome is an ideal landscape to evaluate the occurrence of inverse LGD and the impact of distinct drivers on intraspecific genetic variation in ants.

Study goals and hypotheses

Here, we investigate the distribution of ant intraspecific genetic variation across a latitudinal gradient in Cerrado. We examine biotic and abiotic factors that potentially mediate this genetic diversity, drawing and testing causal hypotheses previously supported by literature. To achieve these goals, we investigated the genetic diversity of a highly abundant Cerrado ant species (*Camponotus crassus* Mayr, 1862) (Figure 1C). This species is one of the most frequent ants on Cerrado foliage (Calixto et al., 2021; Oliveira & Freitas, 2004), commonly feeding on EFNs and trophobiont insects (Del-Claro & Oliveira, 2000; Nogueira et al., 2015; Oliveira & Brandão, 1991), consequently deterring plant herbivores (Oliveira, 1997; Sendoya et al., 2009). These sugar-rich liquid resources account for up to 80% of *C. crassus*' diet (Lange et al., 2019), also mediating *C. crassus* thermoregulatory morphological variation (Azevedo-Silva et al., 2025).

Following the findings that have been consistently reported for distinct levels of ant biodiversity (Dáttilo & Vasconcelos, 2019; Neves et al., 2024; Vasconcelos et al., 2018, 2023), we expected to find an opposite latitudinal gradient for the iGD of *C. crassus*. Given inverse LGDs can arise when higher latitude regions provide more favorable conditions for diversification, such as greater water availability, higher productivity, and higher availability of niches and resources (Cerezer et al., 2022), and as these mechanisms align with well-established drivers of genetic diversity (Pamilo, 1988; Vellend & Geber, 2005), we tested the following three hypotheses:

> **Hypothesis 1.** Genetic diversity of *C. crassus* is affected by environmental descriptors.

Regions with higher energy input and favorable climatic conditions may promote greater diversity at high latitudes, favoring the occurrence of inverse LGD (Cerezer et al., 2022). Mean annual temperature is considered a good predictor of regional energy, consistently correlating with species richness (Cerezer et al., 2022). Also, water availability is considered the most meaningful variable in warm climate regions such as Cerrado (Hawkins et al., 2003), with rainfall being the best predictor of the inverse LGD of ant species in Cerrado (Vasconcelos et al., 2018). Additionally, environmental heterogeneity promoted by vegetation structure strongly influences animal diversity (Tews et al., 2004). Thus, we expect the genetic diversity of *C. crassus* to increase with mean annual temperature, annual precipitation, and vegetation density in Cerrado.

> **Hypothesis 2.** Genetic diversity of *C. crassus* increases with ant community diversity.

Latitudinal gradients of iGD are often positively correlated with species diversity gradients (Gratton et al., 2017; Martin & McKay, 2004; Miraldo et al., 2016; Schär et al., 2017). Because Cerrado ants show an inverse LGD, likely linked to environmental favorability at higher latitudes (Cerezer et al., 2022), we expect a positive influence of ant species richness on *C. crassus* genetic variation.

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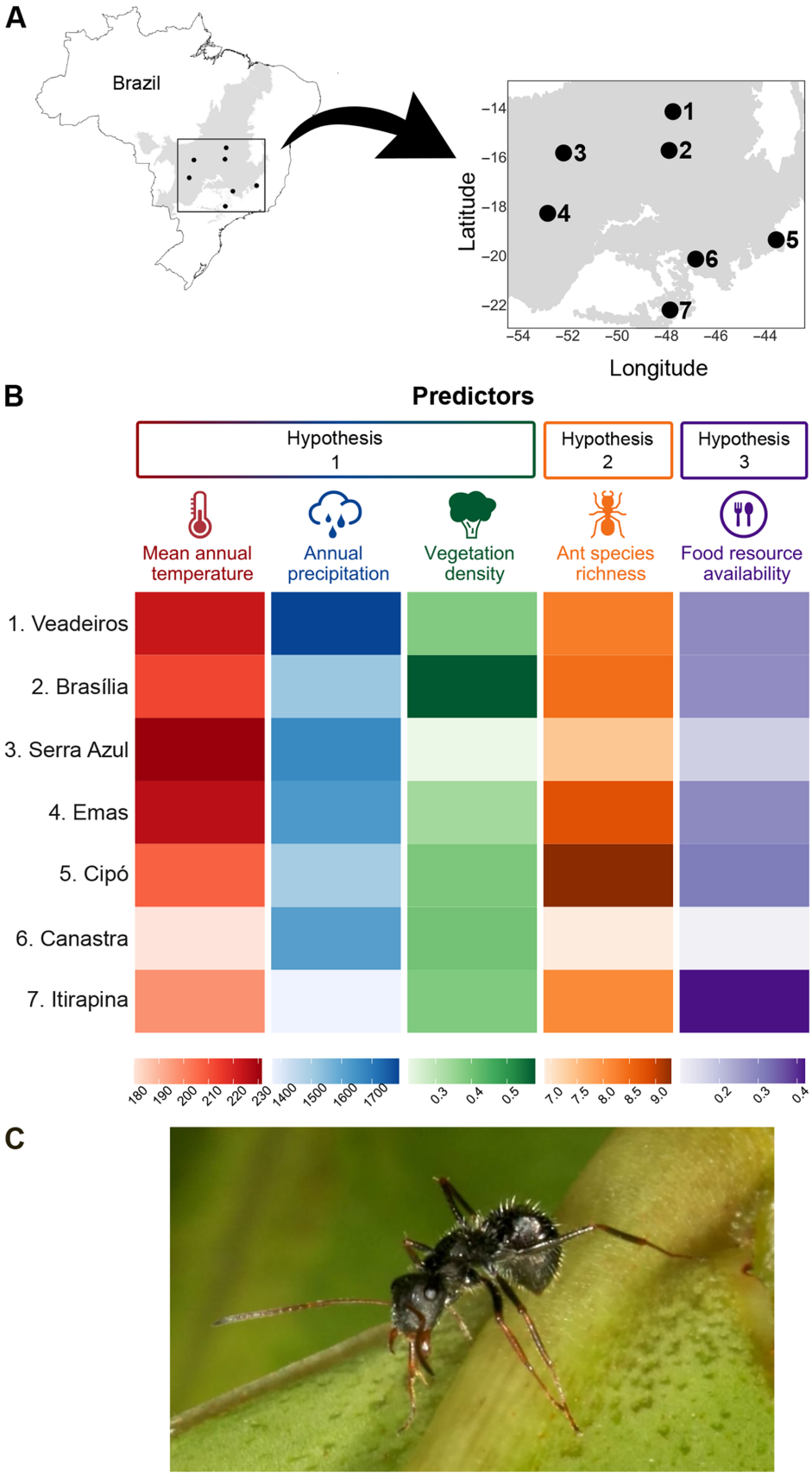


FIGURE 1 (A) Map of Brazil showing the original distribution of the Cerrado (in gray), with sampling sites distribution shown in the large square. (B) Variation of the predictors of *Camponotus crassus* genetic diversity across sites, according to the three initial hypotheses. (C) *C. crassus* worker foraging on Cerrado foliage. Photo credit: S. Sendoya.

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Hypothesis 3. Genetic diversity of *C. crassus* increases with food resource availability.

Sugar-rich resources are central to the ecology of *C. crassus* (Azevedo-Silva et al., 2025; Lange et al., 2019; Oliveira & Brandão, 1991) and can mediate survivorship and reproduction (Tavecchia et al., 2007). Because resource availability is one of the mechanisms thought to generate inverse LGDs (Cerezer et al., 2022), we expect a positive influence of the proportion of plants with EFNs and trophobionts on *C. crassus* genetic variation.

To our knowledge, this is the first study reporting intraspecific genetic distribution and its potential drivers in ants across a latitudinal gradient in a major Neotropical biome.

MATERIALS AND METHODS

Data collection

Sampling strategy

Seven localities distributed across a latitudinal gradient of Cerrado (distant from 173 to 982 km apart; Appendix S1: Table S1) were chosen for data collection (Figure 1A): Estação Ecológica de Itirapina (Itirapina), Parque Nacional Serra da Canastra (Canastra), Parque Nacional Serra do Cipó (Cipó), Parque Nacional das Emas (Emas), Parque Nacional Chapada dos Veadeiros (Veadeiros), Parque Nacional de Brasília (Brasília), and Parque Estadual Serra Azul (Serra Azul). Samplings in all localities were conducted using the same Cerrado physiognomy—Cerrado sensu stricto, a vegetation type characterized by the prevalence of herbaceous vegetation, with trees and shrubs often 3–8 m tall, comprising over 30% crown cover (Oliveira-Filho & Ratter, 2002). Because vegetative growth and EFN activity are higher during the wet season (and ants are therefore more likely to be found on vegetation and ground; Belchior et al., 2012; Silva & Oliveira, 2010), as well as herbivory activity (Marquis et al., 2002), all fieldwork was conducted during southern summer, between November 2012 and April 2013. Observations and collections were carried out during the same period (in the morning) across all localities.

In each sampling site, we established five plots of 200 m, at least 1 km apart from each other. Plots were divided into 20 subplots of 10 m × 2 m, and data were collected in each alternate subplot in the plot, totaling 10 subplots sampled per plot. In each sampled subplot, we considered all small woody plants (diameter at breast height, dbh, of 2–30 mm) up to 1 m from the central line of the plot. We also sampled all large trees (dbh >30 mm) up to 2 m from the central line of the plot. All plants in each subplot were observed during two intervals of 1 min, during which we checked for the presence of (1) trophobiont insects and (2) ants foraging on the plant. Although we observed the presence of EFN during fieldwork, we made extensive search on literature to guarantee presence or absence of EFN on the collected plants. Plants and ants were collected and identified at the species level or classified in morphospecies when identification was impossible. Specimens of *C. crassus* were obtained from samples collected during fieldwork. To reduce the chances of sampling individuals from the same nest, in each plot we selected only one *C. crassus* worker per plant (at least 20 m apart from one another). The total number of *C. crassus* sampled per plot can be found in Appendix S1: Table S2.

Genetic data—Response variables

Using genomic DNA samples obtained with DNeasy Blood & Tissue Kit (Qiagen), we applied a PCR-based approach known as "multiplexed ISSR genotyping by sequencing" (MIG-seq) to discover and genotype single nucleotide polymorphisms (SNPs) (Suyama & Matsuki, 2015). This method reduces genome complexity by amplifying hundreds to thousands of inter-simple-sequence repeats (ISSR). Sequences were treated, filtered, and de novo assembled. Loci were inspected for being putatively under selection (Appendix S1: Figure S1). We recovered 186 SNPs for 143 individuals of *C. crassus*. For full details of genetic data, please see Appendix S1: Section S1. Because the number of sampled workers varied per plot (Appendix S1: Table S2) and genetic estimates can be sensitive to sample size and number of SNPs (Morin et al., 2009), we ran rarefaction analyses of genetic diversity estimates using a minimum sample size of two individuals and considering each sampling site separately. Observed heterozygosity (H_O) and the mean gene diversity within the population (H_S) were both estimated using the function *basic.stats* implemented in the package *hierfstat* (Goudet & Jombart, 2024) in R software v4.2.2 (R Core Team, 2022). Nucleotide diversity (π) was estimated with the function *pi.dosage* in *hierfstat*. For the three genetic diversity estimates, rarefaction results showed that the sample size of seven individuals is enough to recover the genetic variation in *C. crassus* populations with our SNP data (Appendix S1: Figure S2). However, the number of workers sampled in our plots ranged from two to six per plot across the localities (Appendix S1: Table S2). Therefore, we opted for a conservative approach and conducted all the analyses at the site level, pooling all individuals from the same locality

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to reliably estimate the genetic variation of *C. crassus*. Per locality, a total of 23 ants were genotyped for the Estação Ecológica de Itirapina, 22 for the Parque Nacional Serra da Canastra, 19 for the Parque Nacional Serra do Cipó, 18 for the Parque Nacional das Emas, 20 for the Parque Nacional Chapada dos Veadeiros, 23 for the Parque Nacional de Brasília, and 18 for the Parque Estadual Serra Azul. For each locality, *C. crassus* genetic diversity was then characterized by H_O, H_S, and π estimates.

Ecological data—Predictor variables

Because we opted for conducting analyses at site level ($n = 7$), to avoid overfitting and inflated Type I errors, we constrained models to a single predictor variable. Although there is a multitude of variables that can predict ant biodiversity, and due to our limited sampling size, we focused on predictors with clear ecological hypotheses that are also used to explain latitudinal gradients of biodiversity (Cerezer et al., 2022). Given that ecological data were obtained at plot level, and we aimed to perform the analyses across sites at the regional scale, we averaged each predictor variable across plots to have a unique value per site. For latitude and longitude, we estimated a centroid among geographic coordinates of plots from the same site using the function *centroid* implemented in the package *geosphere* (Hijmans, 2024) in R. To characterize environmental descriptors, we estimated mean vegetation density (here calculated as the mean of small and large plant densities) and obtained data of mean annual temperature (bio1) and annual precipitation (bio12) from the WorldClim 2.0 database (Fick & Hijmans, 2017), with a resolution of 30 s. Based on ants sampled on plants during field work, we estimated rarefied ant species richness (using the R package *vegan* – Oksanen et al., 2025) as a *proxy* for ant communities. Finally, food resource availability was estimated as the proportion of plants with sugary resources for ants, which included EFNs and trophobionts.

It is known that genetic variation can be associated with population size (Hartl & Clark, 2010) and species geographic distribution (core populations are predicted to present higher genetic variability than the ones at the edge of distribution; De Kort et al., 2021 and references therein). Thus, we evaluated the influence of population size and species distribution on *C. crassus* genetic diversity. Considering the number of individuals as a direct measure of population size can be problematic for social insects (Gotelli et al., 2011). For instance, the number of ants per plant does not necessarily reflect the number of nearby colonies and can be just an effect of the plant proximity to a single ant colony. Thus, to overcome this issue, we used the proportion of plants with *C. crassus* as a proxy to colony abundance (i.e., population size) in a site.

To evaluate if the genetic variation of *C. crassus* is predicted by its geographic distribution, we conducted species distribution modeling. The occurrence records for *C. crassus* were obtained from the Global Biodiversity Information Facility (GBIF.org, 2025; www.gbif.org) using the package *rgbif* (Chamberlain et al., 2025; Chamberlain & Boettiger, 2017) and complemented with our own plot data. After data cleaning and filtering for spatial accuracy with *Coordinate cleaner* R package (Zizka et al. 2019), we generated a convex hull around all occurrences to define the distributional extent using the package *sp* (Bivand et al., 2013). We computed the centroid of distribution and estimated the distance from our sampling sites to the centroid using the package *geosphere*. Full details of occurrence data processing and *C. crassus* distribution are provided in the Appendix S1: Section S2.

Statistical analysis

We initially assessed Pearson's correlation between the genetic diversity estimates using the *corPlot* function in the R package *psych* (Revelle, 2022). Despite their significant pairwise correlation (Appendix S1: Figure S3A), we decided to include all of them as response variables in downstream analyses. To avoid confusion effects between variables, we also evaluated the correlation between predictors, which were found to be not significantly correlated with each other (Appendix S1: Figure S3B). Thus, all predictors were kept in downstream analyses. We also evaluated the correlations between predictors and latitude, so we could have a better understanding of how predictors change in the geographic space.

We modeled each genetic diversity estimate in response to a single predictor variable using linear models (LM) implemented with the *lm* function in R. As predictors we used: latitude, mean annual temperature, annual precipitation, vegetation density, rarefied ant species richness, the proportion of plants with sugar-rich liquid resources (EFNs and/or trophobionts), the proportion of plants with *C. crassus*, and distance from the centroid of *C. crassus* geographic distribution. Model residuals were inspected for model assumptions using the *simulateResiduals* function implemented in the *DHARMa* package (Hartig, 2025; Appendix S1: Figures S4 and S5). To evaluate the significance of models, we contrasted them to a null model (considering only the intercept) using likelihood ratio tests (LRTs) implemented in the

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lmtest package (Zeileis & Hothorn, 2002). When models were significantly different from the null model, we estimated the standardized effect size of predictors using the *effectsize* function within the *effectsize* package in R (Ben-Shachar et al., 2020) and reported the adjusted R^2 (R^2_{adj}) obtained using the *summary* function of the LM. Finally, model residuals were tested for spatial autocorrelation using Moran I test implemented in the *Moran.I* function of *ape* package (Paradis & Schliep, 2019). All statistical analyses were conducted in R.

RESULTS

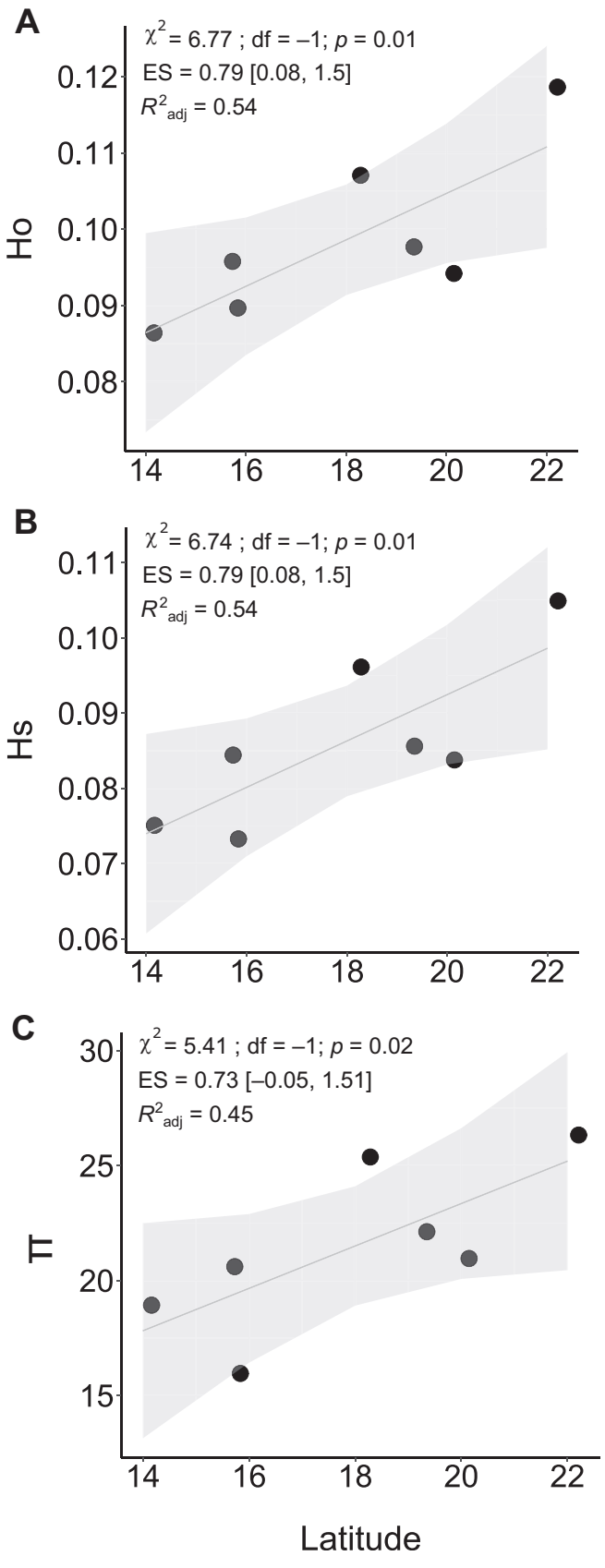
We found a significant reverse latitudinal gradient for all three genetic diversity estimates, with H_O, H_S, and π decreasing toward the equator (Figure 2). Estimates of genetic diversity in response to ecological predictors showed that H_O decreases significantly as annual precipitation increases (LRT; $\chi^2 = 6.4$; $p = 0.01$), with this model explaining a good proportion of the variance ($R^2_{adj} = 0.52$; Figure 3B). A positive association was found between H_O and food resources, with H_O increasing with the proportion of plants with liquid resources (LRT; $\chi^2 = 4.06$; $p = 0.04$, $R^2_{adj} = 0.33$; Figure 3E). Similar results were obtained for estimates of the H_S and marginally significant for π, with both estimates being negatively associated with precipitation and positively correlated with the proportion of plants with sugary resources (Appendix S1: Figure S6). Interestingly, despite the high and moderate correlation between latitude and, respectively, precipitation and sugary resources availability ($r = -0.75$ and $r = 0.25$; Appendix S1: Figure S3B), none of them were significant. Mean annual temperature, vegetation density, ant species richness were not associated with H_O, H_S, and π (Figure 3A,C,D; Appendix S1: Figure S6). Finally, no genetic diversity estimate was significantly correlated with *C. crassus* frequency on vegetation (Appendix S1: Figure S7) or with distance from the core of *C. crassus* geographic distribution (Appendix S1: Figure S8). We found no evidence of spatial autocorrelation for all evaluated models (Appendix S1: Table S3).

DISCUSSION

Our study revealed that an inverse latitudinal gradient is also supported for ant iGD, with *Camponotus crassus* observed heterozygosity (H_O), mean gene diversity within the population (H_S), and nucleotide diversity (π) decreasing significantly at lower latitudes. Our findings corroborate previous studies reporting a reverse latitudinal gradient for other levels of ant biodiversity. Ant species diversity in Cerrado decreases toward the equator for arboreal and ground-dwelling ants, both for habitat-generalist and forest-specialist ants (Vasconcelos et al., 2018), a pattern also found for functional and phylogenetic ant richness (Neves et al., 2024). Similarly, the proportion of the number of plants interacting with ants decreases toward the equator (Azevedo-Silva, 2023), where ant–plant networks also tend to be smaller (Dáttilo & Vasconcelos, 2019). Vertical stratification in Cerrado ant assemblages varies with latitude, with dissimilarity between strata decreasing at higher latitudes (Vasconcelos et al., 2023). Inverse latitudinal gradient in ants is not restricted to Cerrado. Silva and Brandão (2014) found a similar trend in ant species richness in the Brazilian Atlantic rainforest, with ants at higher latitudes presenting less functional differentiation.

Both in Cerrado and Atlantic rainforest, rainfall regimes were commonly found to be the climatic variable underlying latitudinal gradients in ant biodiversity (Azevedo-Silva, 2023; Silva & Brandão, 2014; Vasconcelos et al., 2018). Our analyses of *C. crassus* genetic variation revealed that H_O and H_S are indeed affected by annual precipitation. However, contrary to our initial hypothesis (Hypothesis 1), this is a negative association in which genetic diversity decreases with higher annual rainfall. The explanation for our findings may rely on ant's natural history. In ants, nuptial flights are highly correlated with precipitation, being more frequent after rainy periods (Hölldobler & Wilson, 1990). For some tropical species, nuptial flights are shown to be less common during the dry season, occurring in lower numbers and smaller size in dry forests (Torres et al., 2001). However, for some ant species, mating flights may occur in high frequency under hotter and drier climatic conditions (Boomsma & Leusink, 1981), and rain usually occurs after the flight (Depa, 2006). For these species, the increment in the number of clouds, humidity, and wind velocity can interrupt nuptial flights (Boomsma & Leusink, 1981). Data on *Camponotus crassus* reproductive phenology and breeding system still lack for a better understanding of genetic variation in this species. It is possible that higher precipitation may reduce the period of nuptial flights of *C. crassus* in Cerrado, which would negatively affect mating and, consequently, genetic diversity. But this awaits further investigation. It is also reported that ant polymorphism increases under warm and arid conditions, with such phenotypical diversity being associated with ant persistence in stressful environments (La Richelière et al., 2022). Presumably, the increased genetic diversity of *C. crassus* in areas of low precipitation may facilitate this species to adapt and expand its populations in case the Cerrado becomes hotter and drier under climate change (Hofmann et al., 2021). Given that

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genetic variation is imperative for adaptation to new environmental conditions (Hartl & Clark, 2010), our results suggest that populations at higher latitudes tend to respond better to climate change.

Our results also showed that *C. crassus* genetic diversity (H_O and H_S) increased in areas with a higher proportion of plants with sugary resources, confirming our third hypothesis (Hypothesis 3). Besides accounting for up to 80% of *C. crassus* diet (Lange et al., 2019), sugar-rich resources were also shown to positively mediate variation in pilosity of *C. crassus*, a trait commonly associated with thermoregulation in ants (Azevedo-Silva et al., 2025). Thus, our results show that such food sources are not only important for nutrition and physiology, but also for boosting genetic variation in ant species. Although food resource availability depends on population densities, it can also mediate demography (e.g., Tavecchia et al., 2007). For instance, ant colony growth increases with high resource abundance in the environment (Byk & Del-Claro, 2011; McGlynn et al., 2002). Although we did not detect a significant correlation between population size and genetic diversity, we hypothesize that the abundance of EFNs and honeydew-producing hemipterans in the Cerrado could reduce environmental constraints for *C. crassus*, making it less vulnerable to demographic fluctuations and supporting larger populations with increased genetic variation. In addition to potential impacts on demography, the quality and quantity of resources have also been shown to influence intra and interspecific competition (Grover, 1997; Tilman, 1982; Weider et al., 2008). Under competition, genotypes can be selected depending on resource availability, with superior competitive species often exhibiting higher heterozygosity (Weider et al., 2008). *Camponotus crassus* is highly abundant and dominant ant on Cerrado foliage (Calixto et al., 2021; Oliveira & Brandão, 1991; Oliveira &

FIGURE 2 Linear model for *Camponotus crassus* genetic diversity estimates in response to latitude: (A) Observed heterozygosity (H_O), (B) mean gene diversity within the population (H_S), and (C) nucleotide diversity (π). The likelihood ratio χ^2 statistic, the df, and the corresponding *p* value from the model comparison to the null model using likelihood ratio tests are shown for each model. Standardized effect size (ES) with corresponding CI and adjusted R^2 (R^2_{adj}) are also reported. In each graphic, points represent sampling sites, the solid line represents the model prediction, and the gray bands represents the 95% CIs of each model.

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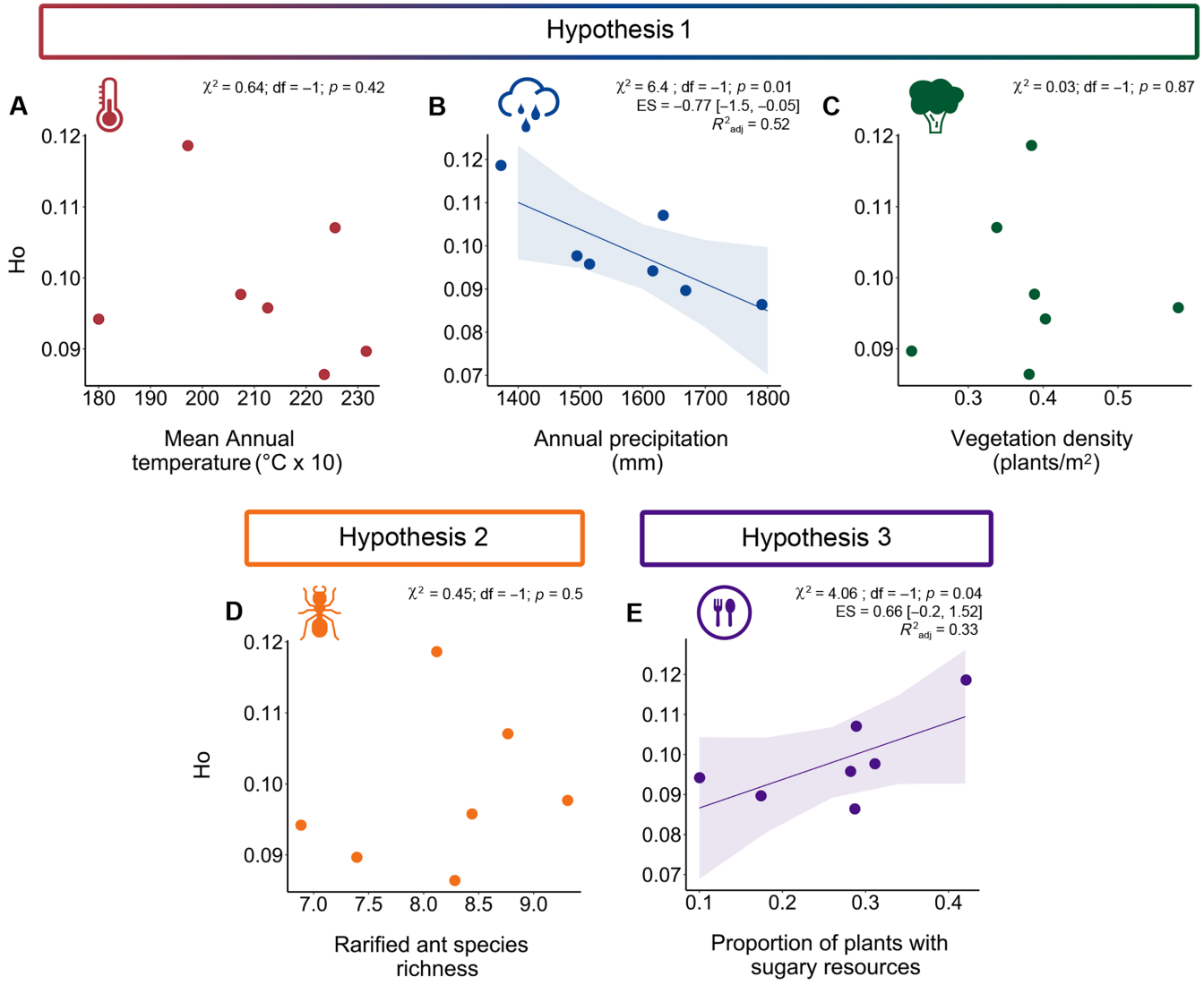


FIGURE 3 Linear models for *Camponotus crassus* observed heterozygosity in response to (A) mean annual temperature, (B) annual precipitation, (C) vegetation density, (D) ant species richness, and (E) food resource availability. Models for mean gene diversity within the population (H_S) and nucleotide diversity (π) can be found in Appendix S1: Figure S6. For each model, the likelihood ratio χ^2 statistic, the df, and the corresponding p value from the model comparison to the null model using likelihood ratio tests (LRT) are shown. Dots represent sampling sites. Effect size (ES) with corresponding CI and adjusted R^2 (R^2_{adj}) are also reported. The solid line represents the model prediction, and the bands represent the 95% CIs of each model. Model prediction and CI are not shown for nonsignificant models.

Freitas, 2004; Schoereder et al., 2010). Thus, it is possible that this species is a superior competitor in ant communities, which would then agree with its higher genetic diversity. We found no significant relationship between *C. crassus* genetic variation and arboreal ant species richness (contrary to the initial expectation of Hypothesis 2). We acknowledge that our sampling strategy was not the most effective method to capture the whole arboreal ant communities (compared to pitfalls, for example), being a *proxy* of the ant species occurrence on plants. Vasconcelos et al. (2018), however, using pitfalls, reported increased ant species diversity at higher latitudes in the Cerrado. Additionally, an inverse latitudinal gradient in ant functional diversity has been observed not only for Cerrado (Neves et al., 2024), but also in the Atlantic rainforest (Silva & Brandão, 2014). Lower functional differentiation between species, combined with an increased species richness at higher latitudes could presumably lead to increased competition within and between species. Under this scenario, new genotypes are more likely to be favored to efficiently exploit liquid food resources and escape competition on foliage (Agashe & Bolnick, 2010; Jousset et al., 2016; Vellend & Geber, 2005). This could be a mechanism explaining latitudinal genetic variations in *C. crassus*. Consequently, the genetically diverse population and high resource competition in higher latitude Cerrado areas could lead to faster rates of evolutionary niche expansion in resource use for *C. crassus* (Agashe & Bolnick, 2010).

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We admit that other variables known to influence ant biodiversity in Cerrado could not be evaluated in this work due to the sampling strategy employed (for instance, climate seasonality, soil fragments, and net primary productivity; Dáttilo & Vasconcelos, 2019; Maravalhas & Vasconcelos, 2020; Neves et al., 2024; Vasconcelos et al., 2018, 2023). Also, other extrinsic and intrinsic factors not evaluated here could potentially influence iGD, such as fecundity, life history traits (De Kort et al., 2021), mutation and recombination rates (Ellegren & Galtier, 2016), individual size (Romiguier et al., 2014), and even phylogenetic history (Leffler et al., 2012). For instance, breeding systems (polygyny, polyandry) have been shown to directly impact the genetic diversity outcomes in *Camponotus* ants in Cerrado (Azevedo-Silva et al., 2020). The genetic diversity of these ants is also mediated by the physiognomic mosaic of the Cerrado, which influences ant dispersal (Azevedo-Silva, Côrtes, et al., 2023). Moreover, habitat fragmentation can also impact ant genetic variation in Cerrado, as demonstrated for *Odontomachus* ants (Ponerinae), in which heterozygosity was found to be higher in colonies at the border of fragments (Azevedo-Silva, Lemos, et al., 2023). Natural history studies of *C. crassus* have been carried out in single locations (e.g., Calixto et al., 2021; Lange et al., 2019). Given that natural history traits can change under distinct environmental conditions (e.g., Pereira-Romeiro et al., 2022), it is possible that fecundity of *C. crassus* varies with location, mediating the observed inverse latitudinal gradient in this species genetic variation. Therefore, we hope our work encourages future research on ant genetic diversity in the Neotropics, expanding the number of populations, ant species, and potential drivers of these levels of biodiversity.

Beyond these Cerrado- and ant-specific processes, our results can also be interpreted in the context of a broader framework on global patterns of genetic diversity. A recent global synthesis demonstrated that population genetic diversity across taxa is shaped by life history traits, contemporary climate, and historical factors such as glacial dynamics and range expansions (De Kort et al., 2021). Particularly, latitude had a small effect, while precipitation and temperature were found to positively correlate with population genetic diversity for distinct vertebrate animal taxa (De Kort et al., 2021). Also, distance from the core of species distribution was a good predictor of genetic variation, with core populations presenting higher genetic diversity than populations at the edge of distributions (De Kort et al., 2021). Contrary to these predictions of the core–periphery hypothesis, we found no clear association between genetic diversity and distance from the center of *C. crassus*' range. Thus, our findings challenge the current emerging view (at least for an invertebrate at the Neotropical region), while aligning to Cerezer et al. (2022) as we reinforce that there is no consensus about the latitudinal gradients, whose patterns and drivers are very taxon-specific and may converge to similar gradients. Besides following an inverse gradient, data on *C. crassus* show that regional climate (mainly precipitation) and resource distributions can mediate genetic diversity. Additionally, the nonsignificant correlation among our predictors and latitude suggests that other ecological and historical factors may override simple spatial expectations, highlighting the importance of integrating species' biogeographic history with local environmental and ecological drivers when explaining patterns of genetic diversity.

Our study casts light on the understanding of ant diversity in Cerrado by documenting evidence of reverse latitudinal gradient for the most fundamental level of biodiversity—genetic diversity—in a dominant ant species. Our large-scale geographic assessment for *Camponotus crassus* fills a gap in the investigation of the drivers of ant iGD. Such an approach is lacking for most ecosystems in dry and warm regions, including the Cerrado savanna.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Data and code (Azevedo Silva et al., 2026) are available from Zenodo: https://doi.org/10.5281/zenodo.15310829. All raw reads of MIG-seq data are available from the National Center for Biotechnology Information (NCBI) Sequence Read Archive (BioProject ID: PRJNA1098407).

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